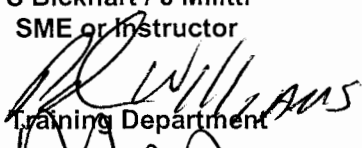
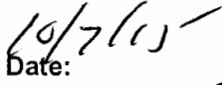
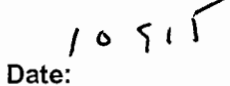


**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

STATION:	SALEM						
SYSTEM:	CVCS						
TASK:	Respond to Loss of VCT level instrument which requires initiating a manual makeup to VCT.						
TASK NUMBER:	0040130101						
JPM NUMBER:	14-01 NRC Sim b						
ALTERNATE PATH:	☒	K/A NUMBER:	APE 022 AA1.01,1.08				
APPLICABILITY:		IMPORTANCE FACTOR:	<table border="1" style="width: 100%; border-collapse: collapse;"><tr><td style="width: 50%; text-align: center;">3.4/3.4</td><td style="width: 50%; text-align: center;">3.3/3.3</td></tr><tr><td style="text-align: center;">RO</td><td style="text-align: center;">SRO</td></tr></table>	3.4/3.4	3.3/3.3	RO	SRO
3.4/3.4	3.3/3.3						
RO	SRO						
	EO ☐	RO ☒	STA ☐ SRO ☒				
EVALUATION SETTING/METHOD:	Simulator						
REFERENCES:	S2.OP-SO.CVC-0006, Rev. 23 (Rev checked all 9-23-15) S2.OP-AR.ZZ-0012, Rev. 37 S2.OP-AB.CVC-0001, Rev. 9 S2.RE-RA.ZZ-0012 Rev. 207 205328-1 rev. 56						
TOOLS AND EQUIPMENT:	None						
VALIDATED JPM COMPLETION TIME:	<u>15 min</u>						
TIME PERIOD IDENTIFIED FOR TIME CRITICAL STEPS:	<u>N/A</u>						
Developed By:	G Gauding Instructor	Date: 8-13-15					
Validated By:	S Bickhart / J Militti SME or Instructor	Date: 9-23-15					
Approved By:	 Training Department	 Date:					
Approved By:	 Operations Department	 Date:					
ACTUAL JPM COMPLETION TIME:							
ACTUAL TIME CRITICAL COMPLETION TIME:							
PERFORMED BY:							
GRADE: ☐ SAT ☐ UNSAT							
REASON, IF UNSATISFACTORY:							
EVALUATOR'S SIGNATURE:		DATE:					

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____

DATE: _____

SYSTEM: CVCS

TASK: Respond to Loss of VCT level instrument which requires initiating a manual makeup to VCT.

TASK NUMBER: 0040130101

INITIAL CONDITIONS: 100% power, MOL. RCS boron concentration is 900 ppm.

SIMULATOR SETUP: IC-242

RT-1 MALF CV0037 VCT LEVEL XMTR LT112 FAILS H/L

Final Value: 100

INITIATING CUE:

You are the Reactor Operator. Respond to all indications and alarms.

Successful Completion Criteria:

1. All critical steps completed.
2. All sequential steps completed in order.
3. All time-critical steps completed within allotted time.
4. JPM completed within validated time. Completion time may exceed the validated time if satisfactory progress is being made (and NRC concurrence is obtained).

Task Standard for Successful Completion:

1. Initiate manual makeup to VCT, and terminate makeup when directed.

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
		Simulator Operator: INSERT RT-1 upon direction of evaluator to fail 2LT-112 high.			
			Announces Control Console alarm VCT Level Hi-Lo as unexpected, and identifies LT-112 has failed high.		
			Refers to S2.OP-AR.ZZ-0012 ARP for console alarm. Note: Operator may enter S2.OP-AB.CVC-0001, Loss of Charging directly.		
	ARP 2.0	Automatic Actions: 2.1 Hi: If 2CV35 is in auto, high level removes normal level control <u>AND</u> shifts 2CV35 fully to HUT.	Notes that 2CV35 3 way valve has automatically shifted to FLOW TO HUT position in response to the high level. Note: Operator may place 2CV35 in MANUAL TO VCT in response to the diversion of letdown flow without an actual high level.		

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
	ARP 3.0	3.1 MONITOR VCT level on LI-112 3.2 <u>IF</u> level is low, <u>THEN</u> : A. INITIATE makeup IAW S2.OP-SO.CVC-0006(Q), Boron Concentration Control. B. ENSURE 2CV35 is lined up to VCT. 3.3 <u>IF</u> level is high, <u>THEN</u> ENSURE 2CV35 directed to HUT. 3.4 <u>IF</u> failure of a VCT level channel has occurred, <u>THEN</u> INITIATE S2.OP-AB.CVC-0001, Loss of Charging.	Monitors VCT level on LI-112. Determines level is not low. Determines actual level is not high (channel has failed high but level is not high) Determines failure of VCT level channel LT-112 has occurred, and initiates S2.OP-AB.CVC-0001, Loss of Charging		
	AB.CVC 3.1	INITIATE Attachment 1, Continuous Action Summary.	Initiates Attachment 1, Continuous Action Summary.		
	AB.CVC 3.2	Is any Charging Pump running?	Determines 23 Charging Pump is running and goes to Step 3.16		
	AB.CVC 3.16	Is there (or has there been) indication of Charging Pump cavitation as evidenced by any of the following: • pump noise • motor amp oscillations • charging flow oscillations	Determines there is no indication of Charging Pump cavitation as evidenced by lack of any of the following: • pump noise • motor amp oscillations • charging flow oscillations and goes to Step 3.54		
	AB.CVC 3.54	Has a Pressurizer level channel failed?	Determines no Pressurizer level channel has failed, and goes to Step 3.62, which continues to 3.63		

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
	AB.CVC 3.63	Has a VCT level instrument (2LT-112 or 2LT-114) failed high (refer to Exhibit 2)?	Determines VCT level instrument 2LT-112 has failed high.		
*	AB.CVC 3.64	TAKE MANUAL control of 2CV35, VCT 3 WAY INLET V, and position it to the VCT.	If not previously performed, takes manual control of 2CV35, VCT 3 WAY INLET V, and position it to the VCT.		
	AB.CVC 3.65	<u>IF</u> letdown is isolated, THEN RESTORE Letdown as follows:	Determines letdown is not isolated.		
	AB.CVC 3.66	<u>IF</u> required to restore VCT level, THEN INITIATE manual makeup to VCT IAW S2.OP-SO.CVC-0006(Q), Boron Concentration Control.	Checks VCT level. Evaluator: <u>If</u> VCT level is NOT <14%, then CUE: The CRS directs you to raise VCT level by 10% from its current level.		
	SO. CVC-6		Performs Prerequisites of S2.OP-SO.CVC-0006, and determines Section 5.2 will be used. Cue: Component Off Normal and Off Normal Tagged review is SAT with no abnormal conditions identified.		
	SO. CVC-6		Reviews Precautions and Limitations section.		
			Operator identifies Section 5.2 Manual Makeup Mode is correct section of procedure to be performed.		

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
*	5.2.1	DETERMINE Boric Acid Flow Setpoint from S2.RE-RA.ZZ-0012, Figure(s), 100A, 100C, and 105 as applicable, <u>OR</u> as calculated and verified by the CRS/STA, <u>AND RECORD</u> Boric Acid Flow Setpoint.	Cue: "BAST boron concentration is 6,700 ppm." <u>IF</u> Operator uses the current makeup setpoint as displayed on the control console, <u>THEN CUE</u> that the CRS directs you to perform Step 5.2.1 to verify the Boric Acid Flow setpoint is correct. Correct Figure to Use is 100A for 62 gpm Primary Water Flow with BAST concentration at normal ppm. 100C is wrong figure because it is for 9000 ppm boron which is not used during normal ops. Figure 105 is the correction factor for RCS temperatures less than 547, which is N/A at 100% power. From the graph, setpoint is slightly less than 10 gpm. Allowable setpoint is 9-11 gpm. Note: If Operator performs calculation using 900 ppm in RCS and 6,700 ppm in BAST, result is 9.62 gpm. Boric Acid setter is in 1 gallon increments.		
	5.2.2	<u>IF</u> required, <u>THEN RESET</u> COUNT A on the Makeup Flow Registers to zero IAW Exhibit 1.	Resets COUNT A for Boric Acid and Primary Water flow IAW Exhibit 1.		
*	5.2.3	PRESS Makeup Control Mode Select STOP pushbutton.	Presses Makeup Control Mode Select STOP pushbutton.		

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
*	5.2.4	PLACE 2CV179, PRIMARY WATER FLOW, in MANUAL, <u>AND</u> CLOSE 2CV179.	Places 2CV179, PRIMARY WATER FLOW, in MANUAL Closes or verifies closed 2CV179 PRIMARY WATER FLOW.		
*	5.2.5	PLACE 2CV172, BORIC ACID FLOW, in MANUAL, <u>AND</u> CLOSE 2CV172.	Places 2CV172, BORIC ACID FLOW, in MANUAL, Closes or verifies closed 2CV172 BORIC ACID FLOW.		
*	5.2.6	ALIGN outlet of Boric Acid Blender to one of the following: A. OPEN 2CV185, MAKEUP FLOWPATH <u>OR</u> B. OPEN 2CV181, MAKEUP FLOWPATH	For one selected flowpath, depresses OPEN pushbutton until OPEN bezel illuminated. The "NOTE" prior to Step 5.2.6 states that charging pump suction is the preferred path, (but not required) which is accomplished by opening the 2CV185. Opening either 2CV185 or 2CV181 is acceptable.		
* *	5.2.7	PERFORM the following as required to support current plant conditions: • START a Primary Water Pump • PLACE a Boric Acid Pump in MANUAL/FAST START	Note: Normal configuration is ONE Boric acid pump running in AUTO SLOW, and NO Primary Water pumps running. Depresses START PB on either PW Pump For selected Boric Acid Pump, depresses MANUAL and FAST START pushbuttons.		
*	5.2.8	ADJUST 2CV172 flow (FI110A) to the value recorded in step 5.2.1	Using INC/DEC pushbuttons, adjusts Boric Acid Flow on FI110A to the value recorded in Step 5.2.1.		

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
	5.2.9	<u>IF</u> required Boric Acid flow is <u>NOT</u> achieved, <u>THEN</u>: • CLOSE 21CV160, RECIRC VALVE • CLOSE 22CV160, RECIRC VALVE •	Closing of the CV160 valves will NOT be required.		
*	5.2.10	Manually ADJUST 2CV179 Setpoint to 62 gpm (or as calculated) to obtain the required flow as indicated on FI111A (Refer to step 3.13)	Using 2CV179 INC/DEC PB's, adjusts PW Flow to 62 GPM +/- 2 GPM as indicated on FI111A. Cue: <u>If</u> Operator cannot get exactly 62 gpm, <u>then cue</u> a range of 61-63 gpm is acceptable. Cue: Once makeup is in progress and operator is monitoring VCT level via LT-114 using the Plant Computer, inform operator that VCT level has risen the required 10%.		
	5.2.11	ENSURE required Boric Acid <u>AND</u> Primary Water Flow are being maintained <u>OR</u> ADJUST 2CV172 and 2CV179 as required to obtain required flows.	Monitors Boric Acid and Primary Water flows being maintained as required and adjusts 2CV172 and 2CV179 as required to obtain required flows.		

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
* * * *	5.2.12	When desired makeup is completed: A. CLOSE the following valves: • 2CV179 • 2CV172 • 2CV185 • 2CV181 B. STOP Primary Water Pump C. PLACE Boric Acid Pump selected in SLOW Speed D. PLACE the system in Automatic Makeup Mode IAW Section 5.1	Depresses the CLOSE pushbuttons for the valves 2CV179 and 2CV172, and whichever of the 2CV185 and 2CV181 was opened in Step 5.2.6. Depresses STOP pushbutton for Primary Water Pump which was started. Places the selected Boric Acid Pump in SLOW speed. Terminate the JPM once the selected Boric Acid Pump has been placed in SLOW speed.		

OPERATIONS DEPARTMENT
JOB PERFORMANCE MEASURE

TQ-AA-106-0303
Revision 4

JOB PERFORMANCE MEASURE VALIDATION CHECKLIST

NOTE: All steps of this checklist should be performed upon initial validation. Prior to JPM usage, revalidate JPM using steps 8 and 11 below.

- QS 1. Task description and number, JPM description and number are identified.
- QS 2. Knowledge and Abilities (K/A) references are included.
- QS 3. Performance location specified. (in-plant, control room, or simulator)
- QS 4. Initial setup conditions are identified.
- QS 5. Initiating and terminating Cues are properly identified.
- QS 6. Task standards identified and verified by SME review.
- QS 7. Critical steps meet the criteria for critical steps and are identified with an asterisk (*).
- QS 8. Verify the procedure referenced by this JPM matches the most current revision of that procedure: Procedure Rev. 23 Date 9/10/2015
- QS 9. Pilot test the JPM:
a. verify Cues both verbal and visual are free of conflict, and
b. ensure performance time is accurate.
- _____ 10. If the JPM cannot be performed as written with proper responses, then revise the JPM.
- _____ 11. When JPM is revalidated, SME or Instructor sign and date JPM cover page.

SME/Instructor: [Signature] B. C. H. H. G.

Date: 9/23/15

SME/Instructor: [Signature] M. I. H.

Date: 9-23-15

SME/Instructor: _____

Date: _____

**OPERATIONS DEPARTMENT
JOB PERFORMANCE MEASURE**

INITIAL CONDITIONS:

1. 100% power, MOL. RCS boron concentration is 900 ppm.

INITIATING CUE:

You are the Reactor Operator. Respond to all indications and alarms.